# B. SUMdamental Decomposition

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1300 | Tags: bitmasks, constructive algorithms, greedy, implementation, math |
https://codeforces.com/problemset/problem/2108/B

On a recent birthday, your best friend Maurice gave you a pair of numbers $n$ and $x$, and asked you to construct an array of **positive** numbers $a$ of length $n$ such that $a_1 \oplus a_2 \oplus \ldots \oplus a_n = x$ $\oplus$.

This task seemed too simple to you, and therefore you decided to give Maurice a return gift by constructing an array among all such arrays that has the smallest sum of its elements. You immediately thought of a suitable array; however, since writing it down turned out to be too time-consuming, Maurice will have to settle for just the sum of its elements.

$\oplus$ denotes the bitwise XOR operation.

## Input

Each test contains multiple test cases. The first line contains the number of test cases $t$ ($1 \le t \le 10^4$). The description of the test cases follows.

Each test case consists of a single line containing a pair of numbers $n$ and $x$ ($1 \le n \le 10^9$, $0 \le x \le 10^9$) — the numbers given to you by Maurice.

## Output

For each test case, output your gift to Maurice — the sum of the elements of the array that satisfies all the described properties. If a suitable array does not exist, output -1.

## Example

| Input |
| --- |
| 8<br>2 1<br>3 6<br>1 0<br>2 0<br>5 0<br>2 27<br>15 43<br>12345678 9101112 |

| Output |
| --- |
| 5<br>8<br>-1<br>2<br>8<br>27<br>55<br>21446778 |

## Note

In the first test case, one of the suitable arrays is [2, 3]. It can be shown that it is impossible to achieve a smaller sum of array elements.

In the second case, one of the suitable arrays is [1, 3, 4]. It can also be shown that this is the optimal amount.